

Task-1

MCQs *(Based on P, NP, NP-Complete, NP-Hard, heuristics, local search, etc.)*

1. What does P stand for?

- A. Polynomial Problem
- B. Polynomial Time
- C. Possible Time
- D. Practical Time

Answer: B) Polynomial Time

2. Problems in class P are:

- A. Hard to verify
- B. Impossible to solve
- C. Efficiently solvable
- D. Always exponential

Answer: C) Efficiently Solvable

3. What is a deterministic algorithm?

- A. Random output
- B. Same input gives same output
- C. Uses guessing
- D. Uses probability

Answer: B) Same input gives same output

4. What does NP stand for?

- A. Non-polynomial
- B. Non-deterministic Polynomial time
- C. New Problem
- D. Numerical Polynomial

Answer: B) Non-deterministic Polynomial time

5. NP problems are:

- A. Easy to solve
- B. Easy to verify
- C. Impossible to verify
- D. Always linear

Answer: B) Easy to verify

6. Which is an example of NP problem?

- A. Binary Search
- B. Merge Sort
- C. Sudoku
- D. Linear Search

Answer: C) Sudoku

7. NP-Complete problems are:

- A. Easy problems
- B. Hardest problems in NP
- C. Only theoretical
- D. Always unsolvable

Answer: B) Hardest Problems in NP

8. If one NP-Complete problem is solved in polynomial time:

- A. All problems become NP
- B. $P = NP$
- C. NP becomes empty
- D. Nothing changes

Answer: B) $P = NP$

9. NP-Hard problems:

- A. Are always in NP
- B. Are easier than NP
- C. May not be in NP
- D. Are always solvable

Answer: C) May not be in NP

10. Decision problems give:

- A. Best solution
- B. Approximate solution
- C. Yes/No answer
- D. Multiple answers

Answer: C) Approximate solution

11. Optimization problems aim to:

- A. Give Yes/No
- B. Find best solution
- C. Random output
- D. Simplify input

Answer: B) Find best solution

12. Heuristic algorithms:

- A. Always give optimal solution
- B. Are slow
- C. Give good enough solutions
- D. Are exact

Answer: C) Give good enough solutions

13. Approximation algorithms:

- A. Have performance guarantee
- B. Are random
- C. Are always exact
- D. Are useless

Answer: A) Have performance guarantee

14. Local search algorithms:

- A. Explore entire search space
- B. Move to neighbor solutions
- C. Always find global optimum

D. Use recursion only

Answer: B) Move to neighbour solutions

15. Hill climbing is:

A. Exhaustive search

B. Greedy local search

C. Divide and conquer

D. Dynamic programming

Answer: B) Greedy local search

MCQs *(Based on exhaustive search, GRASP, complexity, Big-O, etc.)*

1. Exhaustive search is also called:

A. Greedy

B. Dynamic programming

C. Brute force

D. Divide and conquer

Answer: C) Brute force

2. Exhaustive search:

A. Skips solutions

B. Checks all possibilities

C. Uses heuristics

D. Is random

Answer: B) Checks all possibilities

3. Main advantage of exhaustive search:

A. Fast

B. Guaranteed optimal solution

C. Uses less memory

D. Random

Answer: B) Guaranteed optimal solution

4. Exhaustive search complexity is usually:

- A. Linear
- B. Logarithmic
- C. Exponential
- D. Constant

Answer: C) Exponential

5. State space tree represents:

- A. Memory usage
- B. All possible solutions
- C. Sorting steps
- D. Input size

Answer: B) All possible solutions

6. For n items, permutations are:

- A. n
- B. n^2
- C. $n!$
- D. $\log n$

Answer: C) $n!$

7. Circular arrangement formula:

- A. $n!$
- B. $(n-1)!$
- C. n^2
- D. $\log n$

Answer: B) $(n-1)!$

8. GRASP stands for:

- A. Greedy Random Search Process
- B. Greedy Randomized Adaptive Search Procedure
- C. General Random Search Problem
- D. Gradient Random Search

Answer: B) Greedy Randomized Adaptive Search Procedure

9. GRASP combines:

- A. Greedy + Randomization
- B. Divide + Conquer
- C. Recursion + DP
- D. Sorting + Searching

Answer: A) Greedy + Randomization

10. Local search in GRASP is used to:

- A. Generate input
- B. Improve solution
- C. Sort data
- D. Reduce memory

Answer: Improve solution

11. Time complexity measures:

- A. Memory used
- B. Time required
- C. Code size
- D. Output size

Answer: B) Time required

12. Space complexity measures:

- A. Execution time
- B. Memory usage
- C. CPU speed
- D. Input size

Answer: B) Memory usage

13. Big-O notation represents:

- A. Best case
- B. Worst case
- C. Average case

D. Exact case

Answer: B) Worst Case

14. Big Theta (Θ) represents:

A. Upper bound

B. Lower bound

C. Exact bound

D. Random bound

Answer: C) Exact bound

15. Which grows fastest?

A. n

B. n^2

C. $\log n$

D. 2^n

Answer: B) 2^n

Task -2

MCQs Scenario-Based

1.

A program checks every possible combination to solve a problem.
Which technique is used?

A. Greedy

B. Exhaustive search

C. Dynamic programming

D. Heuristic

Answer: B) Exhaustive search

2.

A developer tries all possible routes for delivery and selects the best one.
This is:

- A. Greedy
- B. Exhaustive search
- C. Local search
- D. Approximation

Answer: B) Exhaustive search

3.

A search algorithm guarantees finding a solution if one exists.
Which property is this?

- A. Optimality
- B. Completeness
- C. Efficiency
- D. Randomness

Answer: B) Completeness

4.

An algorithm finds the best solution among all possibilities.
This property is:

- A. Completeness
- B. Optimality
- C. Randomness
- D. Heuristic

Answer: B) Optimality

5.

A system explores all nodes in a decision tree without preference.
This is:

- A. Informed search
- B. Blind search
- C. Greedy search
- D. A* search

Answer: B) Blind search

6.

A problem grows from 10 solutions to millions as input increases.
This is:

- A. Linear growth
- B. Polynomial growth
- C. Combinatorial explosion
- D. Constant growth

Answer: C) Combinatorial explosion

7.

Arranging students in a line where order matters uses:

- A. n^2
- B. $n!$
- C. $(n-1)!$
- D. $\log n$

Answer: B) $n!$

8.

Seating people around a round table uses:

- A. $n!$
- B. $(n-1)!$
- C. n^2
- D. $\log n$

Answer: B) $(n-1)!$

9.

An algorithm builds a solution using greedy choice but adds randomness.
This is:

- A. Dynamic programming
- B. GRASP
- C. Exhaustive search
- D. Divide and conquer

Answer: B) GRASP

10.

A system selects top candidates randomly from a restricted list.
This concept is:

- A. Greedy function
- B. RCL (Restricted Candidate List)
- C. Backtracking
- D. Sorting

Answer:

11.

An algorithm improves a solution by swapping elements repeatedly.
This step is:

- A. Initialization
- B. Local search
- C. Sorting
- D. Recursion

Answer:

12.

You count the number of operations to measure performance.
This is:

- A. Space analysis
- B. Time analysis
- C. Random analysis
- D. Empirical testing

Answer: B

13.

You analyze algorithm performance mathematically without running it.
This is:

- A. A posteriori analysis
- B. A priori analysis
- C. Experimental analysis
- D. Simulation

Answer: B. A priori analysis

14.

An algorithm uses memory depending on input size.
This refers to:

- A. Time complexity
- B. Space complexity
- C. Accuracy
- D. Speed

Answer: B)Space Complexitiy

15.

You compare algorithms ignoring constants and lower-order terms.
This is:

- A. Exact analysis
- B. Asymptotic analysis
- C. Experimental analysis
- D. Random testing

Answer: B. Asymptotic analysis

Case Study 1: Food Delivery Optimization

Q1. Is this a decision problem or optimization problem? Justify.

Ans: Decision problem (it checks whether a route exists within 30 minutes).

Q2. If the company wants the fastest possible route, how does the problem change?

Ans: It becomes an optimization problem.

Q3. Would this problem belong to P, NP, or NP-Hard? Explain your reasoning.

Ans: Decision version $\rightarrow$ NP; Optimization version $\rightarrow$ NP-Hard.

Q4. Suggest one heuristic approach for solving it quickly.

Ans: Greedy approach (choose nearest location first).

Case Study 2: Password Cracking System

Q1. Which type of algorithm is being used?

Ans: Brute-force algorithm.

Q2. What is the time complexity in terms of password length?

Ans: $O(k^n)$.

Q3. Why is this approach not scalable?

Ans: Time grows exponentially with password length.

Q4. Suggest a better approach and explain why it is more efficient.

Ans: Dictionary attack or hashing (reduces search space).

Case Study 3: AI Pathfinding in a Game

Q1. What type of search is being used?

Ans: Informed search (A^*).

Q2. What is a heuristic function, and why is it useful?

Ans: It estimates distance to goal and speeds up search.

Q3. Differentiate between admissible and non-admissible heuristics.

Ans: Admissible never overestimates; non-admissible may overestimate.

Q4. What happens if the heuristic overestimates the cost?

Ans: Optimal solution may not be found.

Case Study 4: University Timetable Scheduling

Q1. Why is this problem considered NP-Hard?

Ans: Large number of combinations with constraints.

Q2. Why is exhaustive search not practical here?

Ans: Exponential time complexity.

Q3. Suggest one metaheuristic method and explain how it helps.

Ans: Simulated annealing (escapes local optima).

Q4. What is the trade-off when using heuristics in this case?

Ans: Faster but not always optimal.

Case Study 5: Hill Climbing Failure

Q1. What problem occurred in the algorithm?

Ans: Stuck in local optimum.

Q2. Explain local optimum vs global optimum.

Ans: Local: best nearby; Global: best overall.

Q3. Suggest two techniques to overcome this issue.

Ans: Random restart, simulated annealing.

Q4. How does simulated annealing improve this situation?

Ans: Allows worse moves to escape local optimum.

Case Study 6: Verification vs Solving

Q1. Why is verification easier than solving?

Ans: Checking is simpler than finding solution.

Q2. Which complexity class does this illustrate?

Ans: NP.

Q3. Give one more real-world example of this concept.

Ans: Checking Sudoku solution.

Case Study 7: Startup Logistics System

Q1. Should they use heuristic or approximation algorithms? Why?

Ans: Heuristic (fast and practical).

Q2. What is the difference between these two approaches?

Ans: Heuristic: no guarantee; Approximation: guaranteed bound.

Q3. Give an example where approximation is better than heuristic.

Ans: Vertex cover approximation.

Case Study 8: Courier Route Planning

Q1. How many possible routes exist? (Write formula)

Ans: $(n-1)! / 2$

Q2. Why is exhaustive search impractical here?

Ans: Factorial growth.

Q3. What is meant by combinatorial explosion?

Ans: Rapid increase in possibilities.

Q4. Suggest a practical alternative approach.

Ans: Genetic algorithm.

Case Study 9: Online Shopping Recommendation

Q1. What type of algorithm is being used?

Ans: Brute-force.

Q2. What are its advantages and disadvantages?

Ans: Accurate but very slow.

Q3. Suggest a more efficient approach and justify.

Ans: Machine learning recommendation system.

Case Study 10: Seating Arrangement Problem

Q1. Which formula should be used: $n!$ or $(n-1)!$? Explain why.

Ans: $(n-1)!$ (circular arrangement).

Q2. How would the result change if seating was in a straight line?

Ans: $n!$

Q3. Why does starting point not matter in circular arrangements?

Ans: Rotation gives same arrangement.

Case Study 11: Algorithm Comparison

Q1. Which algorithm is better for large input and why?

Ans: Algorithm A (linear growth).

Q2. Explain how order of growth is used here.

Ans: Compares scalability of algorithms.

Q3. Why do we ignore constants in complexity analysis?

Ans: They don't affect growth rate.

Case Study 12: Memory Usage Analysis

Q1. What is the total space complexity?

Ans: $O(n)$.

Q2. Differentiate between fixed part and variable part of space.

Ans: Fixed: constant; Variable: depends on input.

Q3. Why is space complexity important in real systems?

Ans: Memory efficiency matters.

Case Study 13: GRASP in Scheduling

Q1. What are the two main phases of GRASP?

Ans: Construction and local search.

Q2. What is the role of the Restricted Candidate List (RCL)?

Ans: Selects good candidates with randomness.

Q3. How does randomness improve the solution quality?

Ans: Avoids local optima.

Q4. Why does GRASP not guarantee optimal solutions?

Ans: It is heuristic-based.

Case Study 14: Big-O Understanding

Q1. What is the Big-O notation?

Ans: $O(n^2)$

Q2. What is the Theta notation?

Ans: $\Theta(n^2)$

Q3. Explain step-by-step how you simplified the function.

Ans: Remove constants and lower terms $\rightarrow$ keep n^2 .

Case Study 15: Blind vs Informed Search

Q1. Identify both approaches

Ans: Blind (A), Informed (B).

Q2. Which is more efficient and why?

Ans: Informed (uses guidance).

Q3. Give real-world examples of both

Ans: Blind: BFS, DFS; Informed: A*, Greedy.